Pavel Vasilyev

Senior AI Engineer - Agentic Systems

Jerusalem, Israel | marchdown@gmail.com | github.com/marchdown | marchdown.hyperstitious.org

PROFILE

Software engineer with a research and applied-mathematics background. Fifteen years of programming, analytics, and systems design, then four years of vibe coding with LLMs.

SKILLS

Foundations: Programming, analytics, systems design, architecture since 2009; research, applied math.

LLM systems: Tool calling, provider routing and fallback, quota and rate-limit handling, cost and latency tracing. Claude, Gemini, OpenRouter, Ollama, Cloudflare Workers AI.

Mathematics and ML: Topological data analysis, quantitative modelling, NLP, feature engineering, model training.

Languages: Rust, Elixir, Python, Swift, Clojure, TypeScript; earlier Haskell and Agda.

Infrastructure: Nix, Caddy, systemd, CI, PWA build and deploy tooling.

EXPERIENCE

Independent Software Engineer - consulting, open source, and LLM-assisted development 2010-present

Consulting and open source since 2010 (150+ projects); LLM coding agents since 2022.

- LLM infrastructure in Elixir, Python, and Rust (private, 2026): provider routing and fallback chains, jittered retry honoring Retry-After, per-client rate limits, token, cost, and latency audit logs. Python harness measured over 88 requests: 100% success on a quota probe, 83% in an outage probe.
- pocket (private, 2026): 18 installable PWAs behind one pipeline. A machine-checked rule set (offline cache, update banner, safe-area, no network on the critical path), content-hash service-worker versions, and a deploy script that installs the Caddy site and verifies the live URLs from outside.
- topo-sonify (public, 2026): Rust library that turns persistent homology, Hodge theory, and Morse theory into sound. Built the mathematical core first and checked it against known invariants ($\beta(S^2) = (1,0,1)$, $\chi = 2$) before writing any audio code; unit-tested topology modules, CI green.
- coggy (private, 2026): Rust cognitive architecture with an AtomSpace, PLN inference, and ECAN attention allocation in a parse, ground, attend, infer, reflect loop; 73 unit tests and CI.

Research Lead - Spring Research 2020-2022

Quantitative research firm focused on financial-market modelling.

- Owned applied research on financial-market data end to end: new signal representations, feature engineering, model selection, and training.

Earlier Data Science, NLP & Software Roles - Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, and Finfon 2009-2019

Roles in Tel Aviv, Hong Kong, and Prague.

- Financial-news NLP and signal extraction; medical-text classification and information extraction; computer-vision pipelines; and corpus-linguistics infrastructure.

PUBLIC WORK (PRIVATE REPOSITORIES ON REQUEST)

topo-sonify - Rust; five written chapters on simplicial complexes, homology, and persistence; CI green.

agda-proofs - Agda formalizations of enumerated and recursive sets and of the natural numbers (2015).

EDUCATION

Education: MSc Computational Linguistics, RSUH, Moscow, 2015. BSc Nuclear Physics, Czech Technical University, Prague, 2009.

Languages: English, Hebrew, Russian; working knowledge of German and French.